The Kenaley Longevity Protocol v5.0 — Physical Pillbox Dispensing Matrix

Daily Chronobiological Timing & Adherence Guide for 7-Day Multi-Compartment Pillboxes | *Rejuvenation Olympics Edition*

(DunedinPACE < 0.45)

SLOT 1: AM FASTED (07:00 – 08:30) | Fasted Mitophagy, Epigenetics & eNOS Tone (Take with 16–24 oz RO Water)

Mod	Intervention	Dose & Form	Core Mechanism & Target	Adherence Note
06	**Urolithin A**	1,000 mg (2 softgels)	PINK1/Parkin fasted mitophagy	Take 100% fasted (0 calories)
07	**Spermidine**	15–20 mg (1–2 caps)	eIF5A Lys50 hypusination / TFEB	Pure trihydrochloride or LKM512
23	**Calcium-AKG**	1,000 mg (2 tabs)	TET1–3 DNA demethylase co-substrate	$\geq$ 24 h away from Fumarate (03)
23	**Liposomal Vit C**	500 mg (1 cap/liq)	Reduces TET catalytic-site $Fe^{3+} \rightarrow Fe^{2+}$	Obligate redox cofactor for TET
23	**Apigenin**	50 mg (1 cap)	CD38 inhibition (preserves NAD^+)	Synergizes with Ca-AKG
10	**Microdose Tadalafil**	2.5–5 mg (1 tab)	Endothelial cGMP/PKG1 $\rightarrow$ eNOS Ser1177	17.5h steady-state half-life
25	**Lithium Orotate**	5 mg elem. (1 cap)	GSK-3β inhibition / BDNF elevation	Microdose (1/100th psychiatric dose)
24	**Cat's Claw (AC-11)**	350–700 mg (1 cap)	PARP Base Excision Repair / NF-κB	Standardized carboxy alkyl esters

Outside Box (AM): Selegiline (22): 0.25 mg *sublingual* (M/W/F only; do not swallow). | **Broth Drink (37):** 15g Collagen + Liposomal C.

SLOT 2: MIDDAY / MEAL 1 (12:00 – 13:30) | Postprandial Vascular, Lipids & ECM Defense

Mod	Intervention	Dose & Form	Core Mechanism & Target	Adherence Note
01	**Acarbose**	25–50 mg (1 tab)	α-Glucosidase inhibitor / Butyrate	**TAKE WITH VERY FIRST FORK**
08	**Bempedoic + Ezetimibe**	180 mg / 10 mg (1 tab)	Hepatic ACLY + NPC1L1 blockade	ApoB < 35 mg/dL / hs-CRP -25%
08	**Citrus Bergamot**	500 mg (1 cap)	AMPK / residual sterol clearance	Standardized polyphenols
09	**Vit K2 (MK-7 all-trans)**	200 μg (1 softgel)	Matrix Gla Protein (MGP) decalcification	Must be 100% *all-trans* isomer
09	**Vit D3 + Retinol**	3,000 IU / 2,500 IU	Retinoid/VDR transcriptional synergy	Balances GGCX expression
11	**Kyolic Garlic (SAC)**	600 mg (1 cap)	Slow-release H_2S persulfidation	Suppresses microbial CutC/CutD
11	**Rhamnan Sulfate**	100 mg (1 cap)	Endothelial heparan sulfate glycocalyx	Sulfated polysaccharide matrix
12	**Antarctic Krill Oil**	1,000 mg (1–2 sgel)	MFSD2A blood-brain barrier LPC-DHA	Marine astaxanthin membrane anchor
13	**Olive Leaf (Oleuropein)**	500 mg (1 cap)	MCU calcium uniporter / AMPK	20% standardized oleuropein
36	**Benfotiamine + Carnosine**	150 mg / 500 mg (1 cap)	Transketolase / MGO dicarbonyl trap	Blocks glucosepane crosslinking
36	**Pyridoxamine (P-5-P)**	50 mg (1 cap)	Amadori intermediate trapping	Prevents ECM stiffening
04	**L-Ergothioneine**	25 mg (1 cap)	OCTN1 IMM radioprotectant	Replaces deprecated Baati C_{60}

SLOT 3: PRE-HORMESIS POUCH (15:30) | Thermal Sauna (28) / Sprinting (38) / HBOT (29) Priming

Mod	Intervention	Dose & Form	Core Mechanism & Target	Adherence Note
27	**Selbex (Teprenone/GGA)**	100 mg (1 cap)	HSF1 co-inducer (3 $\times$ HSP70 in sauna)	Take exactly 60 min pre-sauna
34	**Inorganic Dietary Nitrate**	400 mg (1–2 caps)	Lingual $NO_3^- \rightarrow NO_2^- \rightarrow$ NO vascular surge	Take 90 min pre-workout/sauna

[CRITICAL COLLISION 5 GUARDRAIL]: STRICTLY ZERO ANTIOXIDANTS (No Vit C, PQQ, Astaxanthin, or high-dose polyphenols) in the 4-hour window before/after Sauna, HBOT, or Sprinting!

SLOT 4: EVENING / BEDTIME (20:30 – 22:00) | Membrane Remodeling, Sleep Glymphatics & ECM Cleavage

Mod	Intervention	Dose & Form	Core Mechanism & Target	Adherence Note
35	**DLPC (PPC 40–52%)**	1.2–2.4 g (2–3 sgel)	Tafazzin mature cardiolipin (L_4CL) remodeling	$\geq$ 5 h separated from Krill (12)
35	**TMG (Betaine)**	1.5 g (2 tabs)	BHMT remethylation (spares SAMe for PEMT)	Maintains SAM:SAH ratio > 4.5
11	**Kyolic Garlic (SAC)**	600 mg (1 cap)	Split dose for 24h H_2S endothelial tone	Nightly vascular protection
36	**Chebulic + Rosmarinic Ac.**	500 mg / 250 mg	Cleaves established glucosepane crosslinks	Suppresses Piezo1/YAP-TAZ stiffness
31	**Magnesium L-Threonate**	300–400 mg elem.	GABA-A / SCN cooling / slow-wave sleep	Glycinate/Threonate chelate

STANDALONE DAILY VEHICLES, ATHLETIC PERFORMANCE & SLEEP HARDWARE CHECKLIST

- **10:00 AM Carb-Only Powder Smoothie (37):** 1.5 cups Wild Blueberries + Lemon + Water/Ice. Add: Chondroitin (1.2g), Amla (2.5g), Taurine (2.5g), Glycine (5g), Greens (5g), *Agaricus blazei* (1g), Creatine (5g). (Zero fat/oils for rapid gastric transit).
- **12:30 PM Meal 1 Resistant Starch (37):** High-pressure cooked potatoes/rice chilled at 4°C for 24h (RS3 crystals). Reheat < 60°C + Acarbose 25–50mg.
- **16:00 PM BMT Athletic Protocol (38): Day A** (Warmup plyos → Paused Box Squats → DB Weighted Jumps → RDL hinge) | **Day B** (Sprints → Thrower's Bench → Rows). Intersperse with Broth Drink (15g Collagen + Liposomal C). A/B/A cadence with off-day Zone 2 cardio.
- **20:45 PM Evening Collagen Mugcake (37):** 15g Collagen + Casein + Cacao + Coconut flour (60s microwave). Satiating, non-glycemic nocturnal glycine.
- **22:00 PM Bedtime Hardware (31, 32, 34):** (1) Dissolve *S. salivarius* K12/M18 lozenge on tongue. (2) Topical Tretinoin 0.05% to face. (3) 15-min auricular taVNS (left tragus). (4) 3M Micropore mouth tape horizontally. (5) ERV bedroom ventilation (CO_2 < 750 ppm, 66°F).